

CHESAPEAKE BAY BRIDGE TUNNEL, VA, USA

WMO: 997299

Lat: 37.0365N

Lon: 76.0766W

Elev: 52

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 05-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 2	(b) 23.6	(c) 27.2	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 38.7	(k) 38.1	(l) 36.0	(m) 38.4	(n) 16.5	(o) 320	(p) 0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.6	(c) 88.1	(d) N/A	(e) 86.2	(f) N/A	(g) 84.4	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 14.6	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 40.2	(b) 33.6	(c) 28.7	(e) 20.0	(f) 93.1	(g) 5.5	(h) 2.2	(i) 16.0	(j) 94.6	(k) 12.8	(l) 95.9	(m) 9.7	(n) 97.2	(o) 5.7	(p) 98.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.7	41.2	41.4	47.8	58.2	66.7	75.3	79.0	78.2	73.4	64.5	53.6	48.1
	DBStd	14.98	8.71	7.90	8.04	7.21	6.00	4.86	3.41	3.17	4.30	6.24	6.49	8.22
	HDD50	884	299	258	142	12	0	0	0	0	0	1	37	135
	HDD65	3168	739	659	535	227	53	2	0	0	1	83	343	526
	CDD50	4804	25	19	74	259	519	758	900	874	702	452	146	76
	CDD65	1611	0	0	1	24	107	309	435	409	253	69	2	2
	CDH74	10305	0	0	1	61	321	1969	3629	3112	1091	120	0	1
	CDH80	2018	0	0	0	1	28	404	869	582	132	2	0	0
Wind	WSAvg	14.0	14.7	15.0	15.8	15.1	13.9	12.8	12.2	11.9	13.9	14.5	14.4	14.0
Precipitation	PrecAvg	46.3	3.3	3.0	3.9	3.1	3.5	3.9	5.4	5.1	5.0	3.7	2.9	3.5
	PrecMax	65.2	6.1	9.7	9.4	5.6	5.6	8.9	9.2	11.3	16.9	8.9	9.9	7.5
	PrecMin	31.4	1.4	1.2	0.7	0.7	1.1	1.4	1.1	1.2	0.9	0.1	0.1	1.4
	PrecStd	8.0	1.3	1.8	1.7	1.2	1.1	1.7	2.2	2.8	3.5	2.1	2.1	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.4	66.4	71.2	78.6	83.2	90.2	91.5	89.6	86.8	79.9	71.4	72.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	63.8	61.6	66.7	75.8	80.1	86.7	88.4	87.2	83.8	77.4	68.1	67.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	59.3	57.6	63.7	72.7	77.7	84.5	86.3	85.3	81.0	75.0	65.3	64.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	55.0	53.3	60.4	68.9	75.2	82.1	84.6	83.2	79.2	73.0	62.6	60.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	10.1	10.5	10.4	11.0	9.0	7.9	7.6	6.3	5.8	7.6	8.0	9.5
		MCDBR	16.1	17.2	17.3	16.0	12.7	11.2	10.9	10.0	9.6	9.7	11.9	14.7
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.310	0.320	0.346	0.392	0.435	0.482	0.511	0.505	0.409	0.355	0.325	0.314
	taud		2.547	2.512	2.470	2.354	2.265	2.183	2.116	2.120	2.380	2.525	2.558	2.566
	Ebn at Noon		281	291	292	282	270	257	248	246	267	275	273	272
	Edh at Noon		24	29	33	39	43	47	49	48	36	28	24	23
All-Sky Solar Radiation	RadAvg		733	978	1315	1675	1889	2032	1940	1750	1427	1125	830	653
	RadStd		59	102	103	118	163	132	93	101	115	121	67	52

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (56 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+299	+189	(47)

Nomenclature: See separate page